R-107, First Floor, Metro Tower, Vijay Nagar, Indore (M.P.) 452010

AI-Assisted Workflow Automation — Practical Course Curriculum

Cypher Analytica — Combine automation with AI-powered understanding

This course teaches learners to add AI capabilities—such as classification, extraction, summarisation, and content drafting—to structured business workflows.

Core concept: The workflow defines the sequence and rules; AI performs selected tasks within it. For example, AI classifies an enquiry, while predefined rules decide who receives it.

Duration: 40 hours of guided training + 20 hours of assignments and project work
Audience: Students, business analysts, operations professionals, marketers, and automation developers
Prerequisites: Basic workflow automation, spreadsheet familiarity, and introductory knowledge of APIs and JSON. Coding is optional.

By the end, learners should be able to build AI-assisted workflows with validated outputs, human approval, error handling, and cost monitoring.

Module-wise curriculum

Module	Hours	Topics covered	Practical activity
1. Foundations of AI-Assisted Workflows	3	Traditional automation versus AI-assisted workflows; suitable AI tasks; probabilistic outputs; workflow automation versus autonomous agents; identifying tasks that need human judgment	Redesign a manual enquiry process with clearly defined AI and rule-based steps
2. Connecting AI to a Workflow	4	LLM inputs and outputs; API connection concepts; credentials; model selection criteria; context limits; token usage; separating instructions from incoming data	Connect a workflow to an LLM and summarise a sample message
3. Prompt Design for Business Tasks	4	Task instructions; context; examples; constraints; prompt templates; variable insertion; handling incomplete inputs; versioning prompts	Build reusable prompts for enquiry classification, summarisation, and reply drafting

Module	Hours	Topics covered	Practical activity
4. Structured Outputs and Validation	4	JSON schemas; required fields; permitted categories; data types; parsing; schema validation; missing information; retry and fallback routes	Extract name, course interest, preferred timing, and contact details from an enquiry
5. Classification, Routing, and Prioritisation	4	Intent detection; multi-label classification; routing rules; urgency criteria; ambiguous cases; human escalation; measuring classification quality	Categorise enquiries and complaints, then route them using business rules
6. Document and Email Processing	4	Text extraction; OCR concepts; document segmentation; extracting fields; summarising long content; preserving source references; handling unreadable documents	Extract information from sample application forms and flag fields needing review
7. Knowledge-Grounded Responses	5	Retrieval-augmented generation fundamentals; approved knowledge sources; chunking; embeddings concepts; retrieval; source attribution; outdated or missing information	Draft answers using an approved course FAQ, with references and an escalation path
8. Human Review and Controlled Actions	3	Draft–review–approve processes; approval states; editing generated content; recording decisions; separating generation from external actions	Generate an email draft and send it only after recorded approval
9. Evaluation, Security, and Operations	4	Test datasets; factual accuracy; extraction and routing metrics; prompt injection; sensitive data handling; least-privilege access; logs; retries; latency and cost tracking	Test misleading inputs, invalid outputs, API failures, and duplicate events

Module	Hours	Topics covered	Practical activity
10. Capstone Project and Demonstration	5	End-to-end integration; evaluation; deployment checklist; documentation; monitoring; handover	Demonstrate an AI-assisted enquiry and counselling workflow
Total	**40**		

Practical assignments

1. **AI email summariser:** Produce a concise summary, action items, and unresolved questions.
2. **Enquiry information extractor:** Convert free-text enquiries into validated structured records.
3. **Intelligent enquiry router:** Classify messages and route unclear cases to a reviewer.
4. **Document processing workflow:** Extract application details and highlight missing information.
5. **FAQ-grounded reply generator:** Draft responses using only approved institute information.
6. **Approval-based communication:** Hold AI-generated messages for review before sending.

Capstone: AI-Assisted Admission Enquiry Workflow

Scenario: Cypher Analytica receives enquiries through a form or email. The workflow helps counsellors review and respond efficiently.

Stage	Workflow behaviour
Receive	Capture the enquiry and assign a unique record ID
Validate	Check required fields and detect duplicate submissions
Understand	Use AI to extract course interest, questions, and preferences
Retrieve	Find relevant information from the approved course knowledge base

Stage	Workflow behaviour
Draft	Generate a response supported by the retrieved information
Check	Validate the output and flag unsupported claims or missing details
Approve	Present the draft and source information to a counsellor
Act	Send the approved message and update the enquiry record
Monitor	Log outcomes, usage, failures, and pending follow-ups

The workflow must escalate questions when approved information is unavailable. It must not invent fees, batch dates, eligibility rules, or placement promises.

Required deliverables:

- Process map showing AI steps, rule-based steps, and approval points.
- Working workflow and reusable prompt templates.
- Approved sample knowledge base.
- Evaluation set containing routine, ambiguous, and adversarial inputs.
- Test results, operating guide, and project demonstration.

Use synthetic student information for training.

Assessment and certification

Assessment	Weight
Practical assignments	30%
Output evaluation and troubleshooting	20%
Capstone project	40%
Demonstration and viva	10%